

Team Contract Draft

Project Title: UAV Powertrain Health Monitoring Framework

Team Members:

- Prishsha Chawla – Computer Science (chawlaps@mail.uc.edu)
- Ally Blair – Mechanical Engineering (blaira3@mail.uc.edu)
- Siddharth Urankar – Computer Science (urankasj@mail.uc.edu)

Faculty Advisors:

- Dr. Manish Kumar – University of Cincinnati, Mechanical Engineering
- Dr. Manuel Arias Chao – Zurich University of Applied Sciences & TU Delft
- Dr. Chetan Kulkarni – NASA Ames Research Center
- Luke Busse – PhD Student, University of Cincinnati

Project Focus

This project aims to develop a comprehensive in-flight UAV powertrain health monitoring framework that integrates both semi-supervised and supervised machine learning techniques. Motor health will be assessed through a 1D CNN Autoencoder trained on acoustic signals of BLDC motors, while battery health will be monitored using a Deep Operator Network (DeepONet) to predict degradation, Remaining Useful Life (RUL), and flight time. The ultimate goal is to provide real-time anomaly detection and diagnostics to improve UAV reliability within the In-time Aviation Safety Management System (IASMS).

Roles and Responsibilities

Prishsha Chawla: Development of fault detection models, focusing on the semi-supervised CNN Autoencoder for motor anomaly detection.

Siddharth Urankar: Optimization and integration of models onto the UAV hardware platform, including coding for hardware communication and deployment.

Ally Blair: Mechanical design, drone assembly, sensor/microphone integration, experimental setup, and data collection/testing.

All team members will collaborate on data analysis, presentation preparation, and report writing.

Meeting Schedule & Frequency

Weekly Meetings with Advisors:

- Wednesdays, 9–10 AM (online via Teams) with Dr. Chao, Dr. Kumar, Luke, and students (Dr. Kulkarni will join as available). [May turn into bi-weekly meetings with Dr. Chao and Dr. Kumar and may be joint or separate]

Student Working Sessions:

- Wednesdays, 10 AM–12 Noon (in-person) during Ally's Senior Design discussion section.

- Wednesdays, 12:30–3 PM (in-person at Digital Futures Lab) for UAV build, data collection, and testing.

Monthly Meeting: With Dr. Kulkarni, scheduled as needed.

Additional Meetings: With Dr. Kumar as needed; Luke Busse is available at Digital Futures on Wednesdays for support.

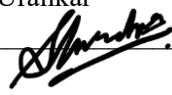
Agreement

By signing this contract, the team members agree to fulfill their responsibilities, communicate regularly, and attend scheduled meetings to ensure the success of the project.

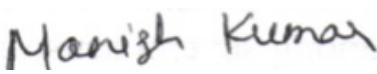
Team Members:

Prishha Chawla 

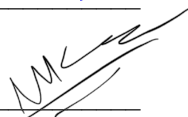
Ally Blair 

Siddharth Urankar 

Faculty Advisors:

Dr. Manish Kumar


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Luke Busse 
